Use Case Specification Table

UC-03: Manage Bookings (Student)

Title	**UC-03 Manage Bookings**
Actor	**Primary:** Student **Secondary:** Email Service
Preconditions	**1. Student is logged into the system.** **2. Student has a valid account with no restrictions.**
Main Flow	**1. Student opens Manage Bookings page.** **2. System displays the main booking interface with available options.** **3. Student selects one of the extended functions:** **- View My Bookings** **- View Tutor Availability** **- Create New Booking** **- Cancel Booking** **4. System processes the selected function according to its specific flow.**

Extended Use Case Flows

A1: View My Bookings

Field	Description
Description	Allows the student to view their existing bookings.
Flow	1. Student selects "View My Bookings". 2. System retrieves all bookings associated with the student. 3. System displays bookings in three tabs: - Upcoming: Confirmed future sessions (status: confirmed) - Past: Completed sessions (status: completed) - Pending: Bookings awaiting tutor confirmation (status: pending) 4. For each booking in Past tab, a "Leave Review" button is displayed.

A2: View Tutor Availability

Field	Description
Description	Allows the student to view a tutor's availability before booking.
Flow	1. Student selects "View Tutor Availability". 2. Student searches for a tutor by skill or name. 3. System displays the selected tutor's profile. 4. System displays the tutor's weekly availability calendar. 5. Student can view available time slots.

A3: Create New Booking

Field	Description
Description	Allows the student to request a new booking with a tutor.
Flow	1. Student selects "Create New Booking". 2. Student searches for a tutor by skill or name. 3. System displays list of matching tutors with their availability. 4. Student selects a tutor and an available time slot. 5. System verifies the time slot is still available. 6. System creates a new booking with status "pending". 7. System <<include>> Send Notification. 8. System displays confirmation: "Booking request sent successfully."

A4: Cancel Booking

Field	Description
Description	Allows the student to cancel an existing booking.
Flow	1. Student selects "Cancel Booking". 2. System displays list of bookings eligible for cancellation based on business rules. 3. Student selects a booking to cancel. 4. Student confirms cancellation. 5. System validates cancellation eligibility again. 6. System changes booking status to "cancelled". 7. System frees the time slot for other students. 8. System <<include>> Send Notification. 9. System displays: "Booking cancelled successfully."

A5: Send Notification

Field	Description
Description	Handles email notifications for booking operations (included in A3 and A4).
Flow	1. System creates email content based on operation type (created/cancelled). 2. System sends notification to Email Service. 3. Email Service delivers email to student's registered address.

Alternative Flows (Exceptions & Errors)

ID	Scenario	Flow
EF-01	Time Slot Unavailable	When creating a booking, if the selected time slot is no longer available (booked by another student), system displays: "Slot no longer available. Please choose another time."
EF-02	Tutor Has No Availability	When viewing tutor availability or creating a booking, if the tutor has not set any availability, system displays: "This tutor has not set any availability yet."
EF-03	No Matching Tutors	When searching for tutors, if no tutors match the skill, system displays: "No tutors found for this skill."
EF-04	No Bookings to Display	When viewing bookings, if the student has no bookings in a specific tab, system displays: "No bookings found."

ID	Scenario	Flow
EF-05	No Cancelable Bookings	When selecting Cancel Booking, if the student has no bookings eligible for cancellation, system displays: "No bookings available for cancellation."
EF-06	Cancel Confirmed Booking Not Allowed	If the student attempts to cancel a confirmed booking outside the allowed cancellation window, system displays: "Confirmed bookings cannot be canceled within 24 hours of the session. Please contact the tutor directly."
EF-07	Cancel Already Cancelled Booking	If the student attempts to cancel a completed booking, system displays: "Cannot cancel a completed session."
EF-08	Cancel Completed Booking	When viewing bookings, if the student has no bookings in a specific tab, system displays: "No bookings found."
EF-09	Email Service Failure	If Send Notification fails to deliver email, system logs the error but does not block the main operation. The booking creation/cancellation still succeeds.

Postconditions

Scenario	Postcondition
View My Bookings	**Booking list displayed with appropriate statuses.**
View Tutor Availability	**Tutor availability calendar displayed.**
Create New Booking	**New booking created with status "pending". Email notification sent.**
Cancel Booking	**Booking status changed to "cancelled". Time slot freed. Email notification sent.**

Relationships

Type	From	To	Description
Association	Student	Manage Bookings	Student initiates the use case
<<extend>>	View My Bookings	Manage Bookings	Optional extended function
<<extend>>	View Tutor Availability	Manage Bookings	Optional extended function
<<extend>>	Create New Booking	Manage Bookings	Optional extended function
<<extend>>	Cancel Booking	Manage Bookings	Optional extended function
<<include>>	Create New Booking	Send Notification	Mandatory part of Create flow
<<include>>	Cancel Booking	Send Notification	Mandatory part of Cancel flow

Use Case Specification Table

UC-06: Manage Tutor Profile (Tutor)

Title	**UC-06: Manage Tutor Profile**
Actor	**Initiated by: Tutor. Involves: Admin (Secondary - for verification)**
Preconditions	1. User is registered in the system as a Tutor. 2. User is logged in.
Main Flow	1. Tutor opens Manage Tutor Profile page. 2. System displays the main profile management interface. 3. Tutor selects one of the extended functions: - Add Skills - Set Availability - Upload Qualifications - Edit Skills - Edit Availability - Delete Qualifications 4. System processes the selected function according to its specific flow.

Extended Use Case Flows

A1: Add Skills

Field	Description
Description	Allows the tutor to add skills to their profile from a predefined list.
Flow	1. Tutor selects "Add Skills". 2. System displays the predefined list of available skills. 3. Tutor selects one or more skills. 4. System verifies each skill is not already added. 5. System adds the skills to the tutor's profile. 6. System displays: "Skill(s) added successfully."

A2: Set Availability

Field	Description
Description	Allows the tutor to set their weekly available time slots.
Flow	1. Tutor selects "Set Availability". 2. System displays a weekly calendar view. 3. Tutor selects days and time slots. 4. System validates the time slots (no overlaps, within business hours). 5. System saves the availability to the database. 6. System displays: "Availability saved successfully."

A3: Upload Qualifications

Field	Description
Description	Allows the tutor to upload qualification files as proof of expertise.
Flow	1. Tutor selects "Upload Qualifications". 2. System displays file upload form. 3. Tutor selects a file from their device. 4. System validates file size (≤5MB) and type (PDF, JPG, PNG). 5. System saves the file with status "pending_verification". 6. Admin will review the file later for verification. 7. System displays: "File uploaded successfully. Pending admin verification."

A4: Edit Skills

Field	Description
Description	Allows the tutor to edit their existing skills.
Flow	1. Tutor selects "Edit Skills". 2. System displays the current list of skills. 3. Tutor adds new skills or removes existing ones. 4. System validates changes. 5. System saves the updated skill list. 6. System displays: "Skills updated successfully."

A5: Edit Availability

Field	Description
Description	Allows the tutor to modify their existing availability.
Flow	1. Tutor selects "Edit Availability". 2. System displays the current weekly availability. 3. Tutor modifies days and time slots. 4. System validates changes. 5. System saves the updated availability. 6. System displays: "Availability updated successfully."

A6: Delete Qualifications

Field	Description
Description	Allows the tutor to delete previously uploaded qualification files.
Flow	1. Tutor selects "Delete Qualifications". 2. System displays the list of uploaded files. 3. Tutor selects a file to delete. 4. System confirms the deletion. 5. System deletes the file from storage. 6. If the file was previously verified, the "Verified" badge is removed from the tutor's profile. 7. System displays: "File deleted successfully."

Alternative Flows (Exceptions & Errors)

ID	Scenario	Flow
EF-01	Duplicate Skill	When adding a skill, if the skill is already in the tutor's profile, system displays: "Skill already added."
EF-02	File Too Large	When uploading a file, if the file size exceeds 5MB, system displays: "File too large. Max 5MB."
EF-03	Invalid File Type	When uploading a file, if the file type is not PDF, JPG, or PNG, system displays: "Only PDF, JPG, PNG allowed."
EF-04	No File Selected	When uploading, if tutor clicks Upload without selecting a file, system displays: "No file selected."
EF-05	Invalid File Extension	When uploading a file without a valid extension, system displays: "Invalid file type."

EF-06	No Skill Selected	When adding skills, if tutor tries to save without selecting any skill, system displays: "Please select at least one skill."
EF-07	Overlapping Time Slots	When setting availability, if tutor selects overlapping time slots, system displays: "Time slots cannot overlap."
EF-08	Invalid Time Range	When setting availability, if tutor selects time outside business hours, system displays: "Please select time within business hours (8:00 AM - 10:00 PM)."
EF-09	File Upload Failure	If the file upload fails due to storage issues, system displays: "Upload failed. Please try again."
EF-10	Delete Non-Existent File	If tutor attempts to delete a file that does not exist, system displays: "File not found."
EF-11	Availability Conflict with Existing Bookings	If tutor tries to remove a time slot that has confirmed bookings, system displays: "Cannot remove this time slot. It has confirmed bookings. Please contact students to reschedule."

Postconditions

Scenario	Postcondition
Add Skills	**Skills added to tutor's profile. Tutor becomes searchable if at least one skill exists.**
Set Availability	**Time slots saved. Students can view them when searching.**
Upload Qualifications	**File saved with status "pending_verification".**
Edit Skills	**Skill list updated.**
Edit Availability	**Availability updated.**
Delete Qualifications	**File removed from storage. Verified badge removed if applicable.**

Relationships

Type	From	To	Description
Association	Tutor	Manage Tutor Profile	Tutor initiates the use case
<<extend>>	Add Skills	Manage Tutor Profile	Optional extended function
<<extend>>	Set Availability	Manage Tutor Profile	Optional extended function
<<extend>>	Upload Qualifications	Manage Tutor Profile	Optional extended function
<<extend>>	Edit Skills	Manage Tutor Profile	Optional extended function
<<extend>>	Edit Availability	Manage Tutor Profile	Optional extended function
<<extend>>	Delete Qualifications	Manage Tutor Profile	Optional extended function